

On an Empirical Model of the Source of Short Wind Waves Based on Radar Observations of Marine Film Slicks

I. Sergievskaya^{ID} and S. Ermakov

Abstract—This study is aimed to develop an empirical model of the spectrum of short wind waves in application to a Bragg component of microwave backscattering from the sea surface at moderate incident angles. The radar backscatter in the model is considered as a composition of Bragg (polarized) and non-Bragg (nonpolarized) components. Data of field experiments on radar probing of film slicks carried out with an X-/C-/S-bands two co-polarized radar instrument were used. An empirical expression for an additional source of Bragg centimeter-scale wind waves is proposed, which made it possible to describe the variations of radar backscatter in slicks when the latter are observed in the cross wind direction. The source is described by a power function of a dimensionless parameter—the wave age. It is shown that the additional source can be considered isotropic, as evidenced by good agreement between the measured contrasts in the upwind direction and those calculated using this model.

Index Terms—Bragg component of microwave scattering, energy balance equation of wind waves, film slicks, radar remote sensing.

I. INTRODUCTION

IT IS well known that microwave backscattering from the sea surface can be used to obtain information about the upper ocean layer and to characterize oil or plastic debris pollutions, nonuniform marine currents, internal waves, areas of wind shadowing, and so on (see [1], [2], [3], [4], [5] and cited literature). However, the problem of characterizing their features using radar sounding has not yet been fully resolved. One of the reasons for this is the limitations of the spectral models of wind waves on the sea surface.

A conventional, widely used model of radar backscattering at moderate incidence angles is a composite model (see [6], [7]). The model states that the normalized co-polarized radar cross section (NRCS) is the sum of a Bragg (polarized) component, determined by the Bragg resonance scattering mechanism, and a nonpolarized (non-Bragg) component determined by longer strongly nonlinear waves, in particular, breaking waves. Based on this assumption, one can find the

Bragg component [8] when measuring NRCS at two co-polarizations—vertical and horizontal (VV and HH), and thus to retrieve the spectral intensity of wind waves at the Bragg wavenumber. The Bragg waves for X- to S-radar bands at moderate incidence angles correspond to gravity-capillary waves (GCWs) of centimeter-scale lengths. The GCW spectrum and, consequently, the Bragg component of microwave backscattering, are very sensitive to the various upper ocean processes mentioned above making the study of Bragg scattering very important.

The hydrodynamic models of the wind wave spectrum are based on the wave energy balance equation accounting for sources, sinks, and nonlinear processes of wind waves. The sources and sinks, such as wind forcing and viscous wave damping, including damping due to surfactant films, have been investigated in detail (see [6], [9], [10], [11], [12]). Weakly nonlinear interactions in various spectral wind wave ranges [13], [14] can serve both as sources and sinks and play an important role in the formation of the equilibrium part of the wind wave spectrum. Strongly nonlinear effects, such as wave breaking, are significant for wind waves of 1 m or less in length and can lead, among other things, to the excitation of centimeter-scale Bragg waves (see [15]). Such excitation can be included in the balance equation as an additional source of short waves [4], [5], which, in particular, must be nonzero for a crosswind direction. This mechanism makes it possible to describe wind waves propagating across the wind, when Phillips's and Miles's wave excitation mechanisms are ineffective, as well as for other directions when, e.g., the viscous wave losses exceed wind forcing. Note that numerous empirical wind wave spectra indicate waves propagating across the wind, which also supports the hypothesis of an additional source in the energy balance equation [16].

This study is focused on an analysis of experimental data on radar probing of the sea surface in the presence of surfactant films using an X-/C-/S-bands two co-polarized radar instrument. Here, the surfactant films are used as a tool to control different parts of the wind wave spectrum. A parameterization of the additional source of centimeter-scale wind waves is proposed based on the experimental crosswind radar contrasts of the Bragg component in film slicks.

II. THEORETICAL BACKGROUND

To retrieve the Bragg component of NRCS, we use the composite model [7], [17] according to which the microwave

Received 9 July 2024; revised 26 August 2024 and 30 September 2024; accepted 30 September 2024. Date of publication 7 October 2024; date of current version 6 November 2024. This work was supported by the Russian Science Foundation (RSF) under Project 23-17-00167. (Corresponding author: I. Sergievskaya.)

The authors are with the Institute of Applied Physics RAS, 603950 Nizhny Novgorod, Russia, and also with the Volga State University of Water Transport, 603950 Nizhny Novgorod, Russia (e-mail: i.sergia@ipfran.ru; stas.ermakov@ipfran.ru).

Digital Object Identifier 10.1109/LGRS.2024.3474796

1558-0571 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

backscattering is a sum of the resonance (Bragg) and the non-Bragg components. The total NRCS $\sigma_{VV,HH}^0$ can be written as follows:

$$\sigma_{pp}^0 = \sigma_{B_pp}^0 + \sigma_{NBC}^0 \quad (1)$$

where $\sigma_{B_pp}^0$ is the Bragg component at VV or HH co-polarizations, and σ_{NBC}^0 is the non-Bragg component. The polarization difference (PD) determined by the wind wave spectrum $F(k_{br})$ at Bragg wavenumber k_{br} is equal to

$$\sigma_{VV}^0 - \sigma_{HH}^0 = 16\pi k_e^4 (g_{VV}^2 - g_{HH}^2)(\theta) F(k_{br}) \quad (2)$$

where $k_{br} = 2k_e \sin \theta$, θ is the incidence angle and k_e is the wavenumber of electromagnetic waves, g_{VV}^2 and g_{HH}^2 are the reflection coefficients, which depend on polarizations of the incident/reflected electromagnetic waves, dielectric constant of the water, and the angle of incidence, as well as on the mean square slopes of long waves [7].

Breaking wave crests significantly contribute to the nonpolarized component, and the breaking events appear in radar return as strong spikes at VV and HH polarizations, as well as in the nonpolarized component [18], [19]. The spikes have nearly the same intensity at VV and HH polarizations but almost absent in PD, i.e., wave breaking weakly contributes to the Bragg component.

A stationary wind wave spectrum can be obtained as a solution to the balance equation for wave action [10], [17], [20] $N(k_{br}, \phi) = F(k_{br}, \phi)\omega(k_{br})/k_{br}$. For the saturation spectrum $B(k_{br}, \phi) = k_{br}^4 F(k_{br}, \phi)$, we have (see [4], [5])

$$\beta_\gamma B(k_{br}, \phi) - \alpha^{n(k_{br})} B(k_{br}, \phi)^{n(k_{br})+1} + I_S(k_{br}, \phi) = 0 \quad (3)$$

where $\beta_\gamma = \beta(u^*, k_{br}, \phi) - 2\gamma(E, \sigma, k_{br})$, $\gamma(E, \sigma, k_{br})$ is the dimensionless wave damping coefficient, which for monomolecular films depends on the elasticity E , σ is the surface tension coefficient, $\beta(u^*, k_{br}, \phi)$ is the dimensionless wind wave growth rate, $u^*(V_{wind})$ is the friction velocity, V_{wind} is the wind velocity, and ϕ is the angle between the wind velocity and wave propagation (or radar look) directions. I_S in (3) denotes the additional source of wind wave generation, and $\alpha(k_{br})$ and $n(k_{br})$ are the parameters of nonlinear limitation of the wave spectrum. Two sources of Bragg waves appear in (3): the first one $\beta(u^*, k_{br}, \phi)$ is due to the instability of wind flow above the water surface (Miles mechanism), the second one (I_S) includes wave excitation by atmospheric pressure pulsations (Phillips mechanism [6]), and generation due to longer nonlinear waves, including wave breaking [5]. The wave energy losses in (3) are described by viscous dissipation $\gamma(E, \sigma, k_{br})$ and by the phenomenological term $\alpha^{n(k_{br})} B(k_{br}, \phi)^{n(k_{br})+1}$. Note that (3) without an additional source was used previously in the local balance model [10], [21].

At low to moderate wind speeds and for the case of a clean water surface, the values of β_γ are normally >0 . In the presence of surfactant films, the sign of β_γ depends on the Bragg wavenumber, radar look direction, surface film characteristics, and the wind speed. If $\beta_\gamma < 0$, the only source of wave generation is the Phillips mechanism and the additional source I_S . For crosswind observations, the wind

growth rate is zero, and thus $\beta_\gamma \equiv -\gamma(E, \sigma, k_{br}) < 0$. In [16] and [22], it was assumed that the amplitudes of Bragg waves propagating across the wind are small, and therefore, the nonlinear term $\alpha^{n(k_{br})} B^{n(k_{br})+1}$ in (3) can be neglected. Under this assumption, the contrast (the ratio of the spectra for clean and film covered surfaces) $K = F_0(k_{br})/F_f(k_{br})$ can be written as $K = K_\gamma = \gamma(E, \sigma, k_{br})/\gamma(0, \sigma, k_{br})$. The K_γ values for not too small film elasticities decrease with wavenumber in the S/C/X ranges and do not depend on wind velocity. This contradicts to the data of field experiments. For instance, according to [16], the contrasts for the cross wind direction depend on wind speed.

III. DATA PROCESSING AND ANALYSIS

Experimental data were obtained during ship experiments on the Gorky Reservoir and experiments from an Oceanographic Platform on the Black Sea in 2018–2022. A microwave coherent dual-polarized S-/C-/X-bands radar developed at IPF RAS was used to measure the radar backscatter at radar wavelengths of 10, 5, and 3 cm and at two (vertical and horizontal) co-polarizations (in more detail, see [3], [16], [18], [19]). A detailed description of the slick experiments is given in [16] and [22]. The incidence angles were about 55° – 60° , and the corresponding Bragg wave numbers were close to 1.1, 2.2, and 3.6 rad/cm for the S-/C-/X-bands, respectively.

To balance the VV and HH channels and avoid errors in estimating PD and PR due to different channel sensitivities, a special calibration procedure was implemented when rotating the radar antenna (a parabolic antenna Rohde and Schwarz AC008) for $\pm 90^\circ$ from its normal position, so that the initial VV channels became the HH ones and vice versa. Calibration was carried out by recording backscattering from wind waves under stable wind conditions, the analysis time in each position was 3–5 min, and the number of rotations was 8–10, which made it possible to accurately estimate the correction factor for each radar band.

Slicks on the water surface were created using a fatty surfactant-oleic acid (OLE) dissolved in ethanol. The average surfactant concentrations on the water surface throughout the experiments exceeded the concentration of the saturated monomolecular film. Such oversaturated films consist of a monomolecular layer with micro droplets of the surfactant and are characterized by practically constant elasticity and surface tension, which for OLE films are 20–25 and 30–35 mN/m, respectively [23].

In our analyses, we selected cases when radar look directions were either upwind or cross wind ones. Practically, taking into account the fluctuations of the wind velocity direction, we considered the cases when variations of the angle ϕ between the look direction and the directions of up- or crosswind did not exceed 20° . We emphasize this fact using the definition of observation direction in brackets as “upwind” or “crosswind.”

The contrasts were determined as the ratio of PD intensities on the clean surface and in the slick area, averaged over time scales much larger than the periods of long wind waves. The wind speed and direction were averaged over the time for which the intensity in the slick area was determined.

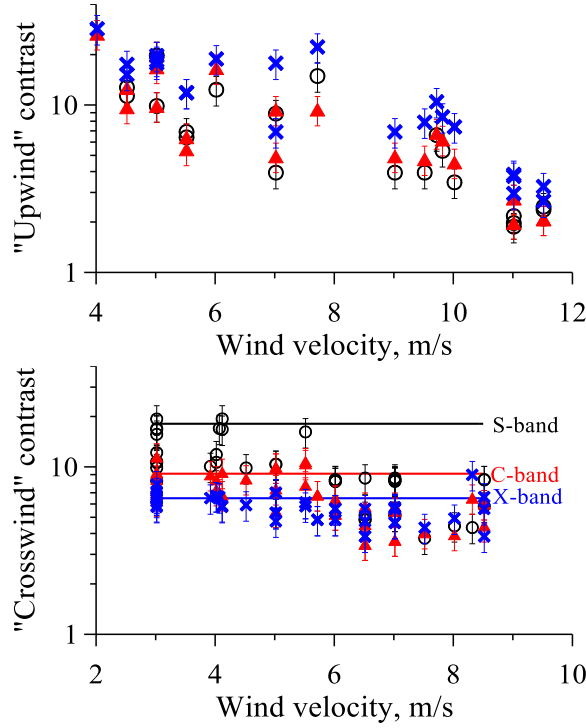


Fig. 1. “Upwind”(top) and “crosswind” (bottom) PD contrast. Black/red/blue (circle/triangle/cross) symbols—S-/C-/X-bands.

Fig. 1 shows the experimental PD contrasts at three wave numbers and two directions of observation. All these contrasts decrease with wind velocity. The “upwind” contrasts (top), in general, increase with wavenumber and are highest in the X-band, while the “crosswind” contrasts (bottom) decrease. The straight lines in Fig. 1(b) correspond to K_γ . The vertical lines indicate the 90% confidence intervals. It is seen that the contrast variations at a given wind velocity are larger than the confidence interval, and this can be due to the variations of wind velocity and direction.

To estimate the intensity of the additional source, we assumed that:

- 1) The additional source has the same intensity on the clean surface and in the film slick areas. This assumption is supported by the relatively weak influence of films on decimeter-meter-scale wind waves, including both suppression of waves shorter than 0.5 m and longer than 1–2 m and amplification in the wavelength range of about 0.5–1 m [24].
- 2) $\alpha(k_{br})$ and $n(k_{br})$ do not depend on film characteristics; their values are presented in [7].

Let us rewrite (3) for the waves of a given wavelength on clean surface and in film slick as follows:

$$\beta_{\gamma,0}\alpha B_0 - \alpha B_0(\alpha B_0)^{n(k_{br})} + \alpha I_s = 0 \quad (4a)$$

$$\beta_{\gamma,f}\alpha B_f - \alpha B_f(\alpha B_f)^{n(k_{br})} + \alpha I_s = 0 \quad (4b)$$

where subscripts “f” and “0” refer to film slick or clean water areas, respectively. From (4), taking into account that the contrast of Bragg waves can be written as $K = F_0/$

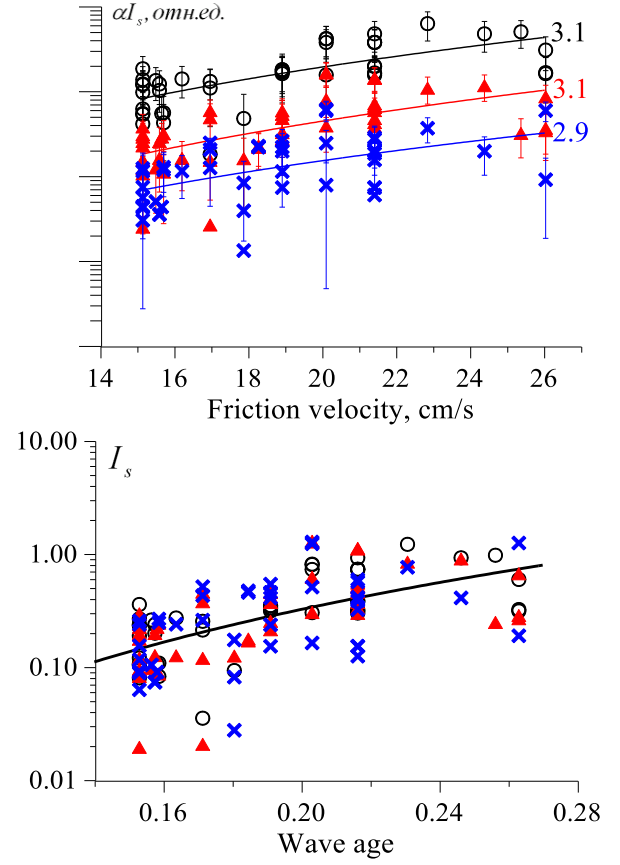


Fig. 2. αI_s versus u^* (top) and $I_s \cdot 10^7$ versus wave age (bottom). “Crosswind” observation. Black/red/blue (circle/triangle/cross) symbols—S-/C-/X-bands. The corresponding color curves are the least square interpolations; the numbers at the curves are the degrees of power functions of the friction velocity.

$F_f = B_0/B_f$, one comes to

$$\begin{aligned} \alpha B_0 &= ((-\beta_{\gamma,f}/K + \beta_{\gamma,0})/(1 - 1/K^{n(k_{br})+1}))^{1/n(k_{br})} \\ \alpha I_s &= -\beta_{\gamma,0}\alpha B_0 + (\alpha B_0)^{n(k_{br})+1}. \end{aligned} \quad (5)$$

Note that the errors of the αB_0 and αI_s estimates increase when the measured contrast is close to β_f/β_0 .

Using the experimental values of contrasts, we can retrieve αB_0 and αI_s , which are determined only by the measured crosswind contrasts and wave damping coefficient, the latter we consider at given parameters of OLE films (see above) as known [23]. We use the following expressions $\beta(u^*, k_{br}, \phi) = C_\beta[k_{br}u^*/\omega(k_{br})]^2 \cos(\phi)$ (see [25]), $\omega(k_{br}) = (gk_{br} + \sigma k_{br}^3)^{1/2}$, $C_\beta = 1.5(\rho_a/\rho_w)(\kappa^{-1} \ln(\pi/k_{br}z_0) - c/u^*)$, ρ_a and ρ_w the air and water densities, $\kappa = 0.4$, $z_0 = a^*u^{*2}/g + a_v v_a/u^*$ is the surface roughness, v_a is the air viscosity, $a_v = 0.1$, and $a^* = 0.018$. The relationship between the friction velocity and wind velocity is taken from [26], $n(k_{br}) = 1/1.2/2.2$ for X-/C-/S-bands, respectively [7]. The dependence of αI_s on friction velocity is shown in Fig. 2 (top). In this figure, the vertical lines indicate errors; some of them for the X-band can be comparable to the mean values. Nevertheless, the tendencies of the intensity increase with friction velocity and the decrease with wavenumber are quite obvious.

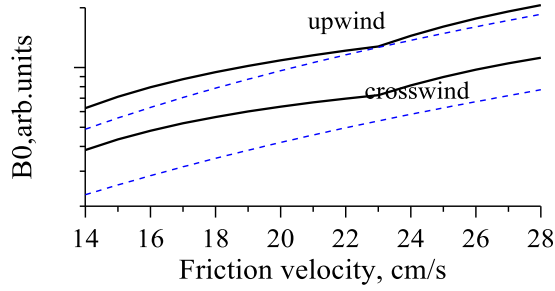


Fig. 3. Saturation spectra at $k_{br} = 2.2$ rad/cm versus friction velocity. Dashed lines—using (5); solid—[27].

IV. DISCUSSION

It follows from Fig. 2 that αI_s can be described by the following dependence on u^* and k_{br} :

$$\alpha I_s \approx 3.4 \cdot 10^{-8} \cdot u^{*3} / k_{br}^2. \quad (6)$$

If we do not consider very low wind speeds and therefore neglect the Phillips mechanism [4], the additional source I_s should be determined mostly by long waves. According to [7], $I_s \propto \beta(u^*, k_{lw}, 0) B(k_{lw}, 0) \propto (u^{*2} k_{lw} / g)^{1+1/n(k_{lw})}$; here, k_{lw} can be assumed as the largest wavenumber for breaking waves, which is about 0.1 rad/cm. Then, we rewrite (6) as $\alpha I_s \propto (u^* \sqrt{k_{lw}/g})^3 \cdot (k_{br}/k_{\gamma})^2$, where $k_{\gamma} = (g/\sigma)^{-1/2}$. The function α can be taken from [7] and for gravity waves and can be written as $\alpha(k_{br}/k_{\gamma}) = (1/a)(k_{br}/k_{\gamma})^{-2} \cdot f$, where $a = 0.002$ and f is a function slowly depending on k_{br} , which can be estimated as 0.9 in the centimeter wavelength range [7], [29]. Then, using (6), we arrive to the following empirical expression for I_s :

$$I_s = b \left[u^* \sqrt{\frac{k_{lw}}{g}} \right]^3 \quad (7)$$

where b is an empirical coefficient $b \approx 5.2 \cdot 10^{-5}$. The dependence I_s on wave age $u^* \sqrt{k_{lw}/g}$ is depicted in Fig. 2 (bottom).

B_0 as a function of wind friction velocity estimated from (5) as well as the spectrum of Elfouhaily et al. [27] are shown in Fig. 3. One can see a very good agreement between two spectra, especially for the upwind direction; for the cross wind direction, the difference between two curves is larger but does not exceed 25%–30%.

Now, we use the empirical (6) based on the data on cross wind contrasts to describe upwind contrasts and to compare with our experiment. The “upwind” contrasts calculated from (4) using the obtained expression (6) for αI_s are presented in Fig. 4, and demonstrate satisfactory agreement with the measured contrasts, at least, at wind velocities below 8–9 m/s. In this velocity range, the variations of experimental data around the theoretical curves do not exceed 10%–15%.

One thus can conclude that the source I_s can be considered as quasi-isotropic. At high wind speeds, the model contrasts are lower than the measured ones, which presumably can be due to wave suppression by turbulence generated by breaking waves [3]. However, this effect was not considered in our analysis because of the lack of knowledge about wave damping due to wave breaking in real marine environment.

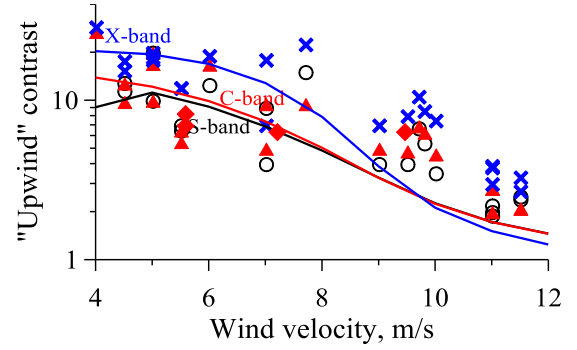


Fig. 4. “Upwind” PD contrasts (curves—model; symbols—experiment; see Fig. 1). Black/red/blue (circle/triangle/cross)—S-/C-/X-bands.

Finally, we compare our empirical expression with the phenomenological formula [28] $I_s = c_b / (2\pi\omega) \iint_{K < K_m} \Omega(K) \beta(u^*, K, \phi) B(K) d \ln K d\phi$, where C_b is a model constant, $K_m = \min(k_{br}/10, k_{lw})$. Supposing the spectrum of long waves in the form $B(K) \sim \beta(u^*, K, \phi)^{1/5}$, the additional source intensity can be estimated as $I_s \propto \omega(K_m) / \omega(k_{br}) (u^* K_m^{1/2} / g^{1/2})^{2.4}$. That is, the source shows a dependence on wind velocity, which is close to the experimental one obtained in this letter.

V. CONCLUSION

An empirical expression is proposed for an additional source of Bragg centimeter-scale wind waves, based on data processing of the radar return from film slicks (films of OLE). The source is described by a power function of a dimensionless parameter—the wave age. It is shown that the additional source can be considered isotropic.

The obtained expression can be used to describe the variability of radar backscattering at the sea surface for solving the problems of detection and identification of surfactant films, plastic debris, variable currents, and so on. It should be emphasized that the presented model is based on the data of special controlled experiments with film slicks. So, when using radar contrasts obtained from the observations of marine slicks and system (5), one can estimate the wave attenuation coefficient in the slicks and thus retrieve the characteristics of marine films (elasticity, thickness, etc.).

In the following work, we plan to validate the obtained expression of the additional source for film slicks of surfactants with other parameters.

ACKNOWLEDGMENT

The authors thank I. Kapustin and A. Kupaev for providing them with raw experimental data and O. Shomina and L. Plotnikov for their participation in the preliminary data processing.

REFERENCES

- [1] V. Klemas, “Remote sensing of ocean internal waves: An overview,” *J. Coastal Res.*, vol. 282, pp. 540–546, May 2012.
- [2] D. Hauser et al., “Satellite remote sensing of surface winds, waves, and currents: Where are we now?” *Surv. Geophys.*, vol. 44, no. 5, pp. 1357–1446, Oct. 2023.
- [3] S. A. Ermakov et al., “Remote sensing of organic films on the water surface using dual Co-polarized ship-based X-/C-/S-band radar and TerraSAR-X,” *Remote Sens.*, vol. 10, no. 7, p. 1097, 2018.

- [4] V. Kudryavtsev and J. Johannessen, "On effect of wavebreaking on short wind waves," *Geophys. Res. Lett.*, vol. 31, no. 20, 2004, Art. no. L20310.
- [5] M. W. Hansen, V. Kudryavtsev, B. Chapron, C. Brekke, and J. A. Johannessen, "Wave breaking in slicks: Impacts on C-Band quad-polarized SAR measurements," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 11, pp. 4929–4940, Nov. 2016.
- [6] O. M. Phillips, "Spectral and statistical properties of the equilibrium range in the wind-generated gravity waves," *J. Fluid Mec.*, vol. 156, pp. 505–553, Jul. 1985.
- [7] V. Kudryavtsev, D. Hauser, G. Caudal, and B. Chapron, "A semi-empirical model of the normalized radar cross-section of the sea surface 1. Background model," *J. Geophys. Res., Oceans*, vol. 108, no. C3, p. 8054, Mar. 2003.
- [8] G. R. Valenzuela, "Theories for the interaction of electromagnetic and oceanic waves—A review," *Boundary-Layer Meteorol.*, vol. 13, nos. 1–4, pp. 61–85, Jan. 1978.
- [9] V. G. Levich, *Physicochemical Hydrodynamics*. Englewood Cliffs, NJ, USA: Prentice-Hall, , 1962.
- [10] M. A. Donelan and W. J. Pierson, "Radar scattering and equilibrium ranges in wind-generated waves with application to scatterometry," *J. Geophys. Res.*, vol. 92, no. C5, pp. 4971–5029, 1987.
- [11] W. J. Plant, "A relationship between wind stress and wave slope," *J. Geophys. Res., Oceans*, vol. 87, no. C3, pp. 1961–1967, Mar. 1982.
- [12] S. A. Ermakov and G. E. Khazanov, "Resonance damping of gravity-capillary waves on water covered with a visco-elastic film of finite thickness: A reappraisal," *Phys. Fluids*, vol. 34, no. 9, p. 10, Sep. 2022.
- [13] M. M. Zaslavski and V. E. Zakharov, "Ranges for generation and dissipation in the kinetic equation for a low-turbulence theory of wind waves," *Izvestiya AS USSR, Atm. Ocean. Phys.*, vol. 18, pp. 821–827, Jan. 1982.
- [14] K. Hasselmann, "On the non-linear energy transfer in a gravity wave spectrum. Part 1: General theory," *J. Fluid Mechanic*, vol. 12, pp. 481–500, Apr. 1962.
- [15] A. Rozenberg and M. Ritter, "Laboratory study of the fine structure of short surface waves due to breaking: Two-directional wave propagation," *J. Geophys. Res., Oceans*, vol. 110, no. C2, pp. 3595–3597, Feb. 2005.
- [16] S. A. Ermakov et al., "New features of Bragg and non-polarized radar backscattering from film slicks on the sea surface," *J. Mar. Sci. Eng.*, vol. 10, no. 9, p. 1262, Sep. 2022.
- [17] O. M. Phillips, "Radar returns from the sea surface—Bragg scattering and breaking waves," *J. Phys. Oceanogr.*, vol. 18, no. 8, pp. 1065–1074, Aug. 1988.
- [18] I. A. Sergievskaya, S. A. Ermakov, A. V. Ermoshkin, I. A. Kapustin, O. V. Shomina, and A. V. Kupaev, "The role of micro breaking of small-scale wind waves in radar backscattering from sea surface," *Remote Sens.*, vol. 12, no. 24, p. 4159, Dec. 2020.
- [19] I. A. Sergievskaya et al., "Modulation of dual-polarized X-band radar backscatter due to long wind waves," *Remote Sens.*, vol. 11, no. 4, p. 423, Feb. 2019.
- [20] B. A. Huges, "The effect of internal waves on surface wind waves. 2. Theoretical analysis," *J. Geophys. Res.*, vol. 83, no. C1, pp. 455–469, 1978.
- [21] S. A. Ermakov et al., "On the effect of surfactant films on changes in the wind-induced waves under the action of internal waves," *Izv. AS USSR, Atmos. Ocean. Phys.*, vol. 16., no. 10, p. 1068, 1980.
- [22] S. Ermakov et al., "Film slicks on the sea surface: Their dynamics and remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 3545–3548, doi: [10.1109/IGARSS39084.2020.9323948](https://doi.org/10.1109/IGARSS39084.2020.9323948).
- [23] S. A. Ermakov and S. V. Kijashko, "Laboratory study of the damping of parametric ripples due to surfactant films," in *Marine Surface Films*. Cham, Switzerland: Springer, 2006, pp. 113–128.
- [24] S. A. Ermakov, "On the intensification of decimeter-range wind waves in film slicks," *Izvestiya, Atmos. Ocean. Phys.*, vol. 46, no. 2, pp. 208–213, Apr. 2010.
- [25] R. W. Stewart, "The air-sea momentum exchange," *Boundary-Layer Meteorol.*, vol. 6, nos. 1–2, pp. 151–167, Mar. 1974.
- [26] S. D. Smith, "Coefficients for sea surface wind stress, heat flux, and wind profiles as a function of wind speed and temperature," *J. Geophys. Res., Oceans*, vol. 93, no. C12, pp. 15467–15472, Dec. 1988.
- [27] T. Elfouhaily, B. Chapron, K. Katsaros, and D. Vandemark, "A unified directional spectrum for long and short wind-driven waves," *J. Geophys. Res., Oceans*, vol. 102, no. C7, pp. 15781–15796, Jul. 1997.
- [28] V. Kudryavtsev, D. Akimov, J. A. Johannessen, and B. Chapron, "On radar imaging of current features. Part 1: Model and comparison with observations," *J. Geophys. Res.* vol. 110, Jul. 2005, Art. no. C07016.
- [29] M. V. Yurovskaya, V. A. Dulov, B. Chapron, and V. N. Kudryavtsev, "Directional short wind wave spectra derived from the sea surface photography," *J. Geophys. Res., Oceans*, vol. 118, no. 9, pp. 4380–4394, Sep. 2013, doi: [10.1002/jgrc.20296](https://doi.org/10.1002/jgrc.20296).